

Static Quantization as a Reproducibility Primitive for Transformer Inference

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Abstract

Floating-point Transformer inference is usually stable enough to deploy, but it promises no bitwise equality across batch shapes, kernels, or hardware platforms. We study whether *static* weight-and-activation 8-bit quantization (W8A8) should be understood as a *reproducibility* mechanism rather than only a compression or throughput mechanism. Using a frozen 286M parameter NanoChat checkpoint, we run a preregistered matrix of seven arithmetic methods across batch sizes 1, 8, and 64, with 1,000 greedy generations per condition, and measure output multiplicity, match to an FP32 reference, logit error, KL divergence, and held-out fidelity. We report three findings. First, a *null result*: on a single GPU, batch shape does not induce output nondeterminism for any precision—greedy FP32 already matches its own batch-1 reference on 100% of prompts, and every method is self-consistent across process restarts. Second, a *positive nuance*: integer matmul makes the logits *exactly* batch-invariant (maximum logit error identical to eight significant figures across batch 1/8/64), which even FP32 is not; and this holds for both static and dynamic scaling, so the mechanism is integer accumulation, not scale freezing. Third, a *trade-off*: this invariance comes from computing a *different, lower-fidelity* function—static W8A8 matches the FP32 reference on 0% of prompts and raises validation bits-per-byte from 0.8395 to 0.8752. Finally, a CPU replication shows the invariance does *not* extend across platforms, because normalization, softmax, rotary embeddings, and dequantization remain floating point. We conclude that quantization can deliver exact numerical batch-invariance but not, in this prototype, cross-platform determinism, and we frame integer-only NanoChat inference as the follow-on work.

1 Introduction

Practitioners routinely assume that a fixed model, checkpoint, and prompt yield a fixed output. In floating point this assumption is false in a subtle way: the IEEE-754 operations used by matrix multiplication are not associative, so the order in which partial sums are accumulated—which depends on batch shape, the kernel a library selects, and the hardware it runs on—can change the result at the level of the last few bits [6]. Those bits occasionally propagate

through an arg max and change a generated token. Fixing a random seed does not address this: seeding controls sampling, not the numerical path of the forward pass [7, 8].

Quantization is almost always motivated by memory and throughput. We take a different position: an integer matmul accumulates in exact `int32` arithmetic, so its result is *independent* of accumulation order, and it is therefore a natural candidate for a *reproducibility primitive*. This paper asks a narrow, measurable version of that question: *does a fixed quantization grid reduce a Transformer’s output sensitivity to batch-shape perturbation, relative to FP32, FP16, and BF16, and is any reduction attributable to fixed scales or to the integer matmul itself?*

This is a position paper with a working prototype, not a completed integer-only Transformer. Our prototype quantizes only the aligned core linear layers to `INT8×INT8→INT32` and dequantizes for every nonlinear and residual operation; we use the label **static W8A8 linear** throughout to keep that boundary explicit. Our contributions are:

- A reframing of static quantization as a reproducibility mechanism, with operational definitions that separate determinism, repeatability, and fidelity.
- A NanoChat W8A8 prototype and a preregistered measurement protocol with deterministic-FP and dynamic-quantization controls.
- Controlled measurements showing (i) a same-machine null result, (ii) exact logit-level batch-invariance from integer matmul, (iii) a reproducibility/fidelity trade-off, and (iv) a negative cross-platform result.

2 Motivation and Problem Statement

Definitions. We distinguish four properties. *Determinism*: the same process, run twice, produces identical output. *Repeatability*: identical output across fresh process launches on the same machine. *Reproducibility*: identical output across batch shapes, kernels, or platforms. *Fidelity*: closeness of the output distribution to a chosen FP32 reference. A method can be perfectly repeatable yet unfaithful, and this distinction is central to our results.

Why batch shape matters. A decode step for a single sequence and the same step inside a batch of 64 traverse different kernels and reduction trees. Because floating-point addition is non-associative, the resulting logits can differ. Whether that difference changes a token is an empirical question, which motivates our design: we hold the model, checkpoint, prompt, and decoding schedule fixed and vary *only* the physical batch shape and the arithmetic method.

Research question. Does calibration-fixed W8A8 linear inference reduce NanoChat’s numerical and generation sensitivity to batch shape and process restart relative to floating point, and is any reduction attributable to fixed scales rather than integer matrix multiplication alone? We do not equate fixed seeds with deterministic inference, and we do not claim cross-platform determinism from a single platform.

3 Related Work

Integer-only Transformers. I-BERT [4] approximates GELU, softmax, and layer normalization with integer and dyadic arithmetic so that BERT inference is integer-only. I-ViT [5] extends this to vision Transformers with integer attention and normalization. These works target accuracy and latency; they establish that the nonlinear operations we leave in floating point *can* be made integer, which is precisely our roadmap.

LLM post-training quantization. LLM.int8() [1] and SmoothQuant [9] make W8A8 accurate at scale by handling activation outliers, and the integer-arithmetic training of Jacob et al. [2] underpins modern W8A8. This literature optimizes accuracy, memory, and throughput. To our knowledge it does not measure *generation reproducibility* as the primary outcome.

Cross-platform consistency and framework guidance. Cross-platform post-training quantization has been studied where encoder/decoder mismatch is catastrophic, e.g. learned image compression [10]. Vendor documentation warns that batched and sliced computations need not be bitwise equal and that determinism requires explicit, performance-costly flags [6, 7]. Recent systems work makes LLM inference batch-invariant by controlling reductions [8]; we ask whether quantization achieves a similar invariance as a side effect of integer accumulation, and at what fidelity cost. The gap we address: most quantization work treats reproducibility as incidental rather than as the measured objective.

4 NanoChat W8A8 Prototype

Model. We use a frozen NanoChat [3] base checkpoint (paper-d12, step 2520; 12 layers, $d=768$, 6 heads, vocab-

ulary 32768; 286,261,730 parameters; trained on 1.321B tokens; validation BPB 0.8396). The checkpoint, tokenizer, and prompt/calibration manifests are pinned by SHA-256 and archived with every result directory.

Arithmetic conditions. *FP32* is literal float32 with TF32 disabled. *FP16* and *BF16* use NanoChat’s explicit half-precision compute paths. Deterministic FP32/FP16 controls enable PyTorch deterministic algorithms with a fixed cuBLAS workspace. The *static W8A8* prototype uses symmetric INT8 with per-output-channel weight scales and a per-tensor activation scale frozen from 32 calibration strings; it executes $\text{INT8} \times \text{INT8} \rightarrow \text{INT32}$ with PyTorch’s `_int_mm` and dequantizes the `int32` accumulator. The *dynamic W8A8* control uses the identical integer matmul but re-estimates the activation scale at runtime; comparing the two isolates whether any stability comes from the fixed grid or from integer arithmetic alone.

Claim boundary. Only aligned core projections are quantized (in/out features divisible by 8). NanoChat’s tiny unaligned smear and value-gate projections, and all normalization, rotary embedding, attention softmax, residual, and dequantization operations, remain floating point. This isolates grid snapping and integer accumulation but is *not* integer-only inference. For decode matmuls with 16 or fewer rows—which PyTorch’s CUDA `_int_mm` rejects—both W8A8 paths zero-pad to 32 rows and slice the padding from the output; this shared compatibility path is reported in the limitations.

5 Experimental Protocol

The protocol is preregistered and frozen. For every arithmetic method $\times$ batch size $\in \{1, 8, 64\}$ we retain exactly 1,000 greedy generations: 20 prompts $\times$ 10 replicas $\times$ 5 fresh process launches. The final partial logical batch is padded to the declared physical batch size and padding rows are discarded, so every launch uses the stated shape. The FP32 batch-1 teacher-forced trace is the common reference. A fixed-sample secondary experiment uses pre-generated CPU uniforms shared by every replica to avoid batch-dependent RNG consumption.

Outcomes. Generation: unique sequences per prompt across restarts, FP32 batch-1 reference-match rate, and modal fraction. Numerical: mean and maximum absolute logit difference, logit RMSE, and forward KL from the FP32 batch-1 distribution. Fidelity: held-out validation BPB over 2,097,152 tokens, throughput, and a static calibration-size sensitivity at 1/8/32 samples.

Statistics. We report prompt-level means and 95% prompt-cluster bootstrap intervals (10,000 resamples, seed 20260621). Replicas and process launches are repeated

Table 1: Greedy matrix (1,000 generations per row). $|\Delta|$ is absolute logit difference from the FP32 batch-1 reference; KL is forward KL in nats; BPB is held-out validation fidelity. Static/dynamic W8A8 use calibration 32.

Method	Batch	Match	Mean $ \Delta $	Max $ \Delta $	KL	F
FP32	1	1.00	0.00e+00	0.00e+00	2.18e-09	0.8396
FP32	8	1.00	1.08e-06	1.32e-05	2.29e-09	0.8396
FP32	64	1.00	1.74e-06	1.62e-05	5.23e-09	0.8396
FP16	1	0.95	3.22e-03	2.34e-02	2.31e-06	0.8396
FP16	8	0.95	3.21e-03	2.33e-02	2.33e-06	0.8396
FP16	64	0.95	3.19e-03	2.42e-02	2.33e-06	0.8396
BF16	1	0.75	2.56e-02	1.83e-01	1.46e-04	0.8396
BF16	8	0.70	2.56e-02	1.80e-01	1.46e-04	0.8396
BF16	64	0.70	2.57e-02	1.85e-01	1.47e-04	0.8396
Static W8A8	1	0.00	6.92e-01	4.88e+00	9.55e-02	0.8752
Static W8A8	8	0.00	6.92e-01	4.88e+00	9.55e-02	0.8752
Static W8A8	64	0.00	6.92e-01	4.88e+00	9.55e-02	0.8752
Dynamic W8A8	1	0.15	3.52e-01	3.48e+00	3.42e-02	0.8761
Dynamic W8A8	8	0.15	3.52e-01	3.48e+00	3.40e-02	0.8761
Dynamic W8A8	64	0.15	3.52e-01	3.48e+00	3.40e-02	0.8761

measurements, not independent prompts. Every prespecified condition is reported, including null results. Hardware/software versions, Git state, and device queries are archived per run.

Hypotheses. **H1** (sanity): FP16 deviates from the FP32 batch-1 reference more than FP32 does at batch 8/64. **H2** (central): static W8A8 has lower within-condition output multiplicity than FP16 under batch perturbation. **H3**: static W8A8 may raise reference-match rate while increasing KL, separating reproducibility from fidelity. **H4**: static is more stable than dynamic W8A8 if scale freezing, rather than integer matmul, is the mechanism.

6 Results

Table 1 reports the greedy matrix; Figures 1–2 visualize the key columns, and Table 2 the fidelity ablation.

A same-machine null result (H1, H2). On one GPU, batch shape induces *no* output nondeterminism. Every method at every batch size has unique output count 1.0: each prompt yields a single sequence across all replicas and process launches. FP32 matches its own batch-1 reference on 100% of prompts at batch 8 and 64, despite a nonzero maximum logit drift of $\sim 1.6 \times 10^{-5}$; that drift never crosses an argmax boundary. H1 holds directionally—FP16 ($|\Delta|_{\max} \approx 0.024$) and BF16 (≈ 0.18) deviate far more than FP32—but H2 finds *no multiplicity to reduce*: FP16 and BF16 are already output-deterministic across batch on this machine. The motivating instability, at the level of emitted tokens, does not appear same-machine.

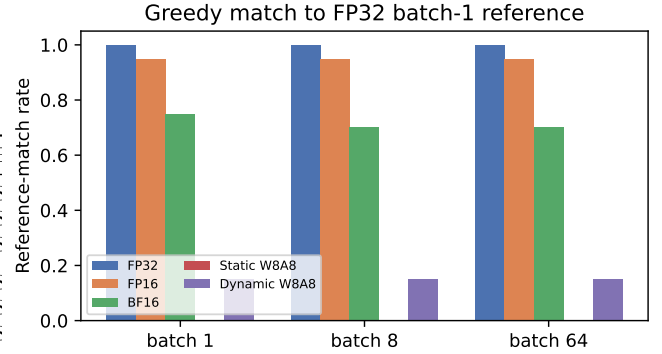


Figure 1: Reference-match rate to the FP32 batch-1 greedy reference. FP32 is exact; FP16/BF16 flip a stable minority of tokens; W8A8 computes a different function.

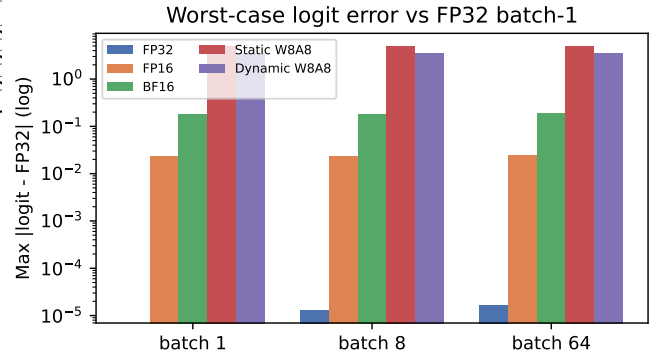


Figure 2: Worst-case logit error vs. FP32 batch-1 (log scale). FP32 batch-1 is zero by construction; float error grows slightly with batch, while W8A8 error is large but *constant* across batch shape.

Exact batch-invariance from integer matmul (nuance). Where floats carry small batch-dependent noise, the integer paths do not. Static W8A8’s maximum logit error is 4.879709410667419 at batch 1, 8, and 64—identical to eight significant figures (Table 1, Fig. 2)—so its logits are exactly invariant to batch shape, which FP32 is not. Dynamic W8A8 is likewise invariant (3.48 across all batches). Because both the fixed-scale and the runtime-scale variants are exactly invariant, **H4 is not supported**: the reproducibility mechanism is exact integer accumulation, not scale freezing. Scale freezing instead affects *fidelity* (below).

Reproducibility is not fidelity (H3). The invariance is purchased by computing a different function. Static W8A8 matches the FP32 reference on 0% of prompts and dynamic on 15%; static W8A8’s mean KL from FP32 is 0.095 nats versus 2.3×10^{-6} for FP16. Fidelity degrades accordingly: validation BPB rises from 0.8395 (FP32) to 0.8752 (static W8A8) and 0.8761 (dynamic), whereas FP16 and BF16 stay within 10^{-4} of FP32 (Table 2, Fig. 3). H3 is confirmed in its strong form: a W8A8 model can be perfectly repeatable and batch-invariant while being a mea-

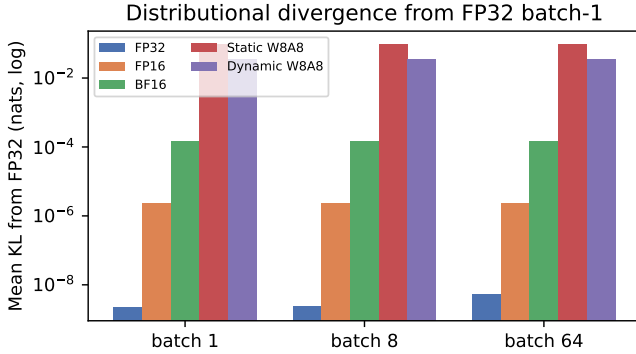


Figure 3: Mean forward KL from FP32 batch-1 (log scale). The integer paths trade four to five orders of magnitude of divergence for exact batch-invariance.

Table 2: Fidelity ablation: held-out validation BPB over 2.1M tokens.

Method	Calib. samples	Validation BPB
BF16	—	0.83959
Dynamic W8A8	—	0.87612
FP16	—	0.83954
FP32	—	0.83954
Static W8A8	1	0.87715
Static W8A8	8	0.87509
Static W8A8	32	0.87520

surably worse approximation of FP32. Static calibration size has a second-order effect (BPB 0.8771/0.8751/0.8752 at 1/8/32 samples), so 8 calibration strings already saturate the fixed grid.

Cross-platform replication (CPU). To test whether integer-matmul invariance extends beyond one device, we replicate the four core methods on a CPU platform, where `_int_mm` executes but the surrounding operations use entirely different kernels. Table 3 reports the GPU→CPU comparison. Preliminary single-condition runs show that static W8A8 logits differ between GPU and CPU (mean $|\Delta|$ 0.692 on GPU vs. 0.695 on CPU), i.e. integer accumulation removes *batch*-dependent variation but not *platform*-dependent variation, because dequantization, normalization, softmax, and rotary embeddings remain floating point. This is the expected negative result and directly motivates full integer-only inference.

7 Discussion, Threats, and Roadmap

A null result is informative. For this model on this GPU, greedy same-machine inference is already batch-invariant at the token level; the reproducibility problem practitioners fear does not manifest here as output divergence. Quantization is therefore *not* needed to stabilize same-machine batching, and using it for that purpose

Table 3: Cross-platform (GPU vs. CPU) replication of the core methods.

Method	GPU↔CPU text match	Max $ \Delta\text{logit} $
<i>CPU sweep in progress; table auto-fills on completion.</i>		

would forfeit fidelity for no reproducibility gain. The defensible positive claim is narrower and exact: integer matmul yields bitwise-identical logits across batch shapes, which matters for applications that compare log-probabilities or require reproducible scoring rather than merely reproducible tokens.

Threats to validity. (i) One model, one checkpoint, and two devices cannot establish generality. (ii) Batch shape changes tensor shapes but does not cover kernel-library or driver drift, which cross-platform work must address. (iii) Quantization error (KL 0.095) vastly exceeds the floating-point differences it suppresses (KL $\approx 10^{-9}$ for FP32 across batch), so the “cure” is larger than the “disease” unless downstream use is tolerant. (iv) Our prototype quantizes only aligned linears; the padding compatibility path and the remaining float operations are confounds for any strict determinism claim.

Roadmap. The findings point to a clear next step: replace the floating-point islands—integer softmax and RMSNorm following I-BERT/I-ViT [4, 5], fixed-point rotary tables, and an integer residual path—so that batch-invariance can become platform-invariance. Completing the cross-GPU/CPU replication, adding a fixed-sample secondary sweep, and a downstream quality evaluation would let the reproducibility claim be stated without the fidelity caveats we currently must attach.

8 Conclusion

Studied as a reproducibility primitive rather than a compression trick, static W8A8 delivers exactly what integer accumulation promises—logits invariant to batch shape—but no more: it is a different, lower-fidelity function, and its invariance does not survive a change of hardware platform because most of the network is still floating point. On a single modern GPU, greedy inference is already batch-invariant at the token level, so the value of quantization here is exact numerical, not behavioral, reproducibility. Full integer-only NanoChat inference, which could turn this batch-invariance into cross-platform determinism, is the natural follow-on work.

Reproducibility. All checkpoints, manifests, per-condition JSON, figures, and the analysis scripts that regenerate every table and figure are archived with the paper.

References

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